

Appendix J - Full Memory Map

Every region of the Intuition Engine address space, in address order, with its size and purpose. The map is identical from the IE64, IE32, M68K, and x86 sides; the 6502 and Z80 see windowed views into it (Chapters 26 and 27).

J.1 Low RAM and stacks

Range	Size	Purpose
0x000000-0x0003FF	1 KB	M68K vector table; x86 IVT; trap vector base for IE64.
0x000400-0x022FFF	~140 KB	BASIC ROM image, system call shim, monitor stub.
0x023000-0x04FFFF	180 KB	BASIC program text (BASIC_PROG_START-).
0x050000-0x057FFF	32 KB	BASIC simple variables.
0x058000-0x05FFFF	32 KB	BASIC string variables.
0x060000-0x08BFFF	176 KB	BASIC arrays.
0x08C000-0x08FFFF	16 KB	BASIC string temporaries.
0x090000-0x096FFF	28 KB	BASIC GOSUB / FOR stack.
0x097000-0x09EFFF	32 KB	Free RAM (general user).
0x09F000-0x09FFFF	4 KB	IE32 stack (STACK_START).

J.2 PC-compatible VRAM apertures

Range	Size	Purpose
0x0A0000-0x0AFFFF	64 KB	VGA graphics window (mode 13h, mode 12h linear).
0x0B0000-0x0B7FFF	32 KB	Reserved (PC-compatible monochrome text window; not implemented).
0x0B8000-0x0BFFFF	32 KB	VGA text buffer (0xB8000).

J.3 General RAM gap

Range	Size	Purpose
0x0C0000-0x0CFFFF	64 KB	Free RAM (general user).
0x0D0000-0x0DFFFF	64 KB	Voodoo texture RAM.
0x0E0000-0x0EFFFF	64 KB	Free RAM (general user).

J.4 The MMIO region (0xF0000-0xFFFFF)

Range	Size	Device
0xF0000-0xF049B	1180 B	Video chip + palette + extended blitter.

Range	Size	Device
0xF0700-0xF07FF	256 B	Terminal / serial / input.
0xF0800-0xF0B7F	896 B	SoundChip.
0xF0BC0-0xF0BD7	24 B	MOD player.
0xF0BD8-0xF0BEB	20 B	WAV player.
0xF0C00-0xF0C20	33 B	PSG / AY.
0xF0C30-0xF0C37	8 B	SN76489.
0xF0C40-0xF0CFF	192 B	SID2 flex block.
0xF0D00-0xF0D20	33 B	POKEY.
0xF0D40-0xF0DFF	192 B	SID3 flex block.
0xF0E00-0xF0E2D	45 B	SID.
0xF0E80-0xF0EFF	128 B	SFX triggers.
0xF0F00-0xF0F5F	96 B	TED audio + video.
0xF1000-0xF13FF	1 KB	VGA registers.
0xF1400-0xF140F	16 B	HOST appliance block.
0xF2000-0xF2017	24 B	ULA registers.
0xF2100-0xF213F	64 B	ANTIC.
0xF2140-0xF21B7	120 B	GTIA.
0xF2200-0xF221F	32 B	File I/O.
0xF2260-0xF22AF	80 B	Amiga Paula DMA.
0xF2300-0xF231F	32 B	Media loader.
0xF2320-0xF233F	32 B	Image executor.
0xF2340-0xF238F	80 B	Coprocessor.
0xF2390-0xF23AF	32 B	Clipboard bridge.
0xF23B0-0xF23BF	16 B	Coprocessor extended monitor.
0xF23C0-0xF23DF	32 B	IRQ diagnostics.
0xF23E0-0xF23FF	32 B	Bootstrap loader.
0xF2400-0xF24FF	256 B	SysInfo (RAM-size ABI).
0xF8000-0xF87FF	2 KB	Voodoo 3D registers.
0xFA000-0xFBAFF	6912 B	ULA VRAM aperture.

J.5 Main video RAM

Range	Size	Purpose
0x100000-0x5FFFFFFF	5 MB	Main VRAM (video chip framebuffer, 1920x1080 RGBA).

J.6 High RAM and reserved

Range	Size	Purpose
0x600000-0xEFFFFFFF	9 MB	Free RAM (general user / coprocessor heap).
0xF00000-0xFEFFFF	960 KB	Reserved (high-MMIO mirror window).
0xFF0000-0xFFFFFFFF	64 KB	Reserved (M68K sign-extended low-word aliases).

J.7 The 6502 / Z80 views

The 6502 sees a 16-bit address space 0x0000-0xFFFF. Its adapter routes the high page-window region as follows:

6502 range	Maps to (32-bit)
\$0000-\$00FF	zero page in main RAM.
\$0100-\$01FF	6502 stack page in main RAM.
\$0200-\$CFFF	main RAM.
\$D000-\$D7FF	per-chip 6502-style apertures (PSG, SID, POKEY, TED, VGA, ULA).
\$D800-\$DFFF	ULA paged VRAM data port.
\$E000-\$EFFF	bank-selectable window into main RAM.
\$F000-\$FFF9	MMIO mirror of 0xF0000-0xF0FF9, with \$F700-\$F705 and \$F7F0 intercepted as bank registers.
\$FFFA-\$FFFF	NMI / reset / IRQ vectors.

The Z80 sees the same shape with port I/O substituted for the chip-aperture region; the bank-window scheme is identical.

J.8 Notes

The map above is the engineering view: every byte of the 24-bit bus that has a defined meaning. Most user programs use a small sub-set of it - the BASIC variable area, the I/O region, and the main video RAM - and never touch the rest. Programs that compete for memory (a large in-memory dataset, a custom blitter routine, a coprocessor worker that wants its own heap) should carve their storage out of the "Free RAM" gaps in J.1, J.3, and J.6 rather than colliding with the BASIC variable region.